Exos Aerospace | Recoverable µg Cryo-Transfer Flight Demo

SBIR 2026.R6 | DAF26BX06-DV513 (D2P2)

Retiring the on-orbit refuel function: tank-to-tank cryogenic propellant transfer in microgravity on a reusable suborbital platform

Autonomous On-Orbit Logistics & Sustainment



BLK3 reusable VTVL vehicle, tethered hover test, Greenville TX

Concept of Operations

1 **Ground 1-g transfer baseline (Greenville test set)**

2 **Integrate in BLK3 payload bay**

3 **µg coast: tank-to-tank cryo transfer, instrumented**

4 **Recover · inspect · refly**

Refuel is the gating function of an on-orbit logistics enterprise. Exos flies the transfer hardware in microgravity, brings it back, and flies it again.

Objective

- Demonstrate **tank-to-tank cryogenic propellant transfer in microgravity** (mass gauging, thermal, pressure, flow) and compare it against a characterized 1-g ground baseline.
- Deliver recovered flight hardware and a flight dataset that de-risks the **refuel** leg of DV513's resupply / refuel / repair / recover architecture.

Innovation

- **Recoverable, repeatable** space-environment test points for a fraction of the cost and schedule of an orbital campaign.
- Linerless composite **Type-V cryo tankage** (the class that flew as fuel and LOX tanks on IM-1 Odysseus and IM-2 Athena) is the leading tank candidate.

Team

- **Exos** (prime / integrator, BLK3 flight + Greenville transfer test set) · **Princeton Cryogenics** (transfer system, cryo-thermal) · **Scorpius Space Launch Co.** (Type-V tankage, subject to trade).
- **Lockheed Martin (proposed):** active Phase II teammate. Receiver-side / architecture requirements, data review, and the Phase III transition path.

Ask of LM: Letter of Support + Phase II teaming on DV513.

Cost and Schedule ($K, 12-month D2P2)

Element	$K
Exos labor: PM, systems eng., integration, test	355
Princeton Cryogenics: transfer system & cryo-thermal	185
Source/receiver tank set (Type-V candidate)	130
Transfer hardware, QD, instrumentation	95
Ground transfer campaign (Greenville)	80
Flight integration + suborbital demo ops	95
Travel, reserve, G&A	45
Total (12 months)	**985**

Milestones: M1 SRR + cryogen/surrogate decision (Mo 2) · M2 PDR + tank down-select (Mo 4) · M3 CDR (Mo 7) · M4 ground 1-g transfer campaign (Mo 9) · M5 µg flight transfer + recovery (Mo 12) · final report + transition package

Proposal-stage estimate. TABA (if allowed) is additive. LM Phase II scope/share to be set jointly.

Success & Transition

Success: an instrumented µg cryogenic transfer, recovered and inspected, with a flight-vs-1-g dataset and a transition package delivered.

1 **D2P2 (DV513)** — recoverable µg cryo-transfer flight demo, LM on the team — **$985K / 12 mo**

2 **Phase III with LM** — transfer + tankage scaled into LM on-orbit logistics / depot architectures — **LM-integrated**

3 **USSF sustainment** — refuel capability for LEO/MEO/GEO/cislunar assets; repeat flights qualify upgrades — **SSC / SpaceWERX**

4 **Commercial OTV** — Type-V tank sets already baseline tankage for a commercial OTV program — **dual-use pull**

Distribution: non-proprietary summary; no export-controlled data (no LM NDA in place). Nothing is awarded. BLK3 received its FAA launch license Mar 2026 and is preparing for flight; Exos has flown commercial payloads since 2018 (SARGE). Type-V tanks are flight-proven as tankage; in-space cryo transfer on this architecture has not been demonstrated and is the objective of this effort.

Exos × Lockheed Martin | DAF26BX06-DV513

Backup: why Exos, and what we are asking of Lockheed Martin

Why Exos

• One of only three FAA-licensed vertical-takeoff reusable rocket operators with demonstrated recovery/reflight (with SpaceX and Blue Origin).

• Commercial payloads flown since 2018 (SARGE). BLK3 is our latest FAA-licensed vehicle (Mar 2026).

• Minimal infrastructure: needs a pad plus power/comms. Fast, recoverable, repeatable test cadence.

• Greenville propellant-transfer ground test set for the 1-g baseline before any flight.

• Current USAF SBIR Phase II performer (FA9300-25-C-6022).

Tankage heritage

• Scorpius linerless composite Type-V cryo tanks flew as fuel and LOX tanks on IM-1 Odysseus and IM-2 Athena (TRL 9 as tankage).

• Exos holds an 18% stake in Scorpius; Type-V tanks are a leading candidate, subject to a fit/cost/schedule trade.

Proposed LM role (Phase II)

• Define receiver-side and interface requirements from LM on-orbit logistics / servicing architectures.

• Participate in design reviews (SRR / PDR / CDR) and flight data review.

• Co-own the Phase III transition plan into LM programs and USSF sustainment.

Ask

• Letter of Support for the DV513 D2P2 submission.

• Confirmation of LM Phase II teaming intent; Exos will report the government award outcome to LM.

• An NDA if LM wants technical detail beyond this non-proprietary summary.

Point of contact

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